UI-TARS-2 Technical Report: Advancing GUI Agent with Multi-Turn Reinforcement Learning

ByteDance Seed

Paper version: arXiv 2509.02544v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Open problems remain in data scalability, multi-turn reinforcement learning (RL), the limitations of GUI-only operation, and environment stability.*

A **native** agent model needs data, a stable sandbox and a hybrid action space.

Motivation: three open gaps

1. **Scarce data.** Long GUI trajectories are costly.
2. **Unstable RL.** Sparse rewards over long sequences.
3. **GUI-only limits.** Files and terminals matter.

Evaluation setup

OSWorld · 369 tasks | WindowsAgentArena · 150+ | AndroidWorld · 116 | Online-Mind2Web · 300 | 23B active, 230B MoE | GUI or SDK actions

- **All online;** no offline step accuracy.
- **Held out:** not specified.

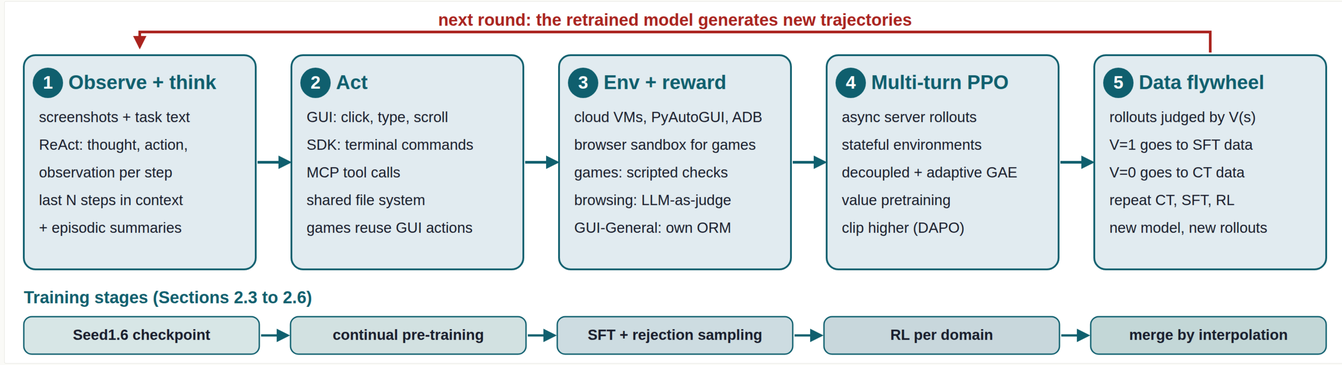
Our read (CUA Radar)

67	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; strong baselines; RL-vs-SFT and GUI-vs-SDK; judge test; held-out partial.
- **Flags:** AP4, learned judge: ORM F1 83.8 on 300 human-labelled traces; misclassified cases show a relatively high false-positive rate.
- Pattern C11. Seeds, matched budget, cost: not reported. Authors' numbers, not rerun.

Method: one native model, a hybrid sandbox and a data flywheel

- **Sees / acts:** screenshots in; click, type, scroll, terminal and MCP calls out.
- **New:** hybrid GUI + SDK sandbox, data flywheel, and a stabilised multi-turn PPO recipe.
- **Signal:** verifiable outcome reward: scripted checks, LLM judge, or its own ORM.



Our redraw (not the paper's figure): the ReAct loop, sandbox, PPO recipe and data flywheel from Section 2.

$$P(t_n, a_n \mid \text{instruction}, W_n, o_n, E_n)$$

Next thought and action from working memory W and episodic memory E

Key results: online success rate (%)

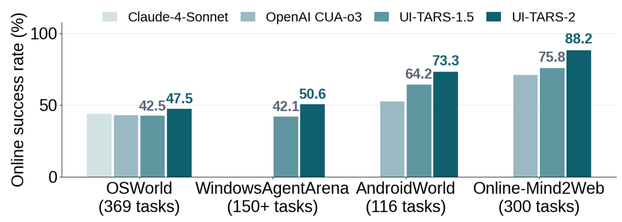
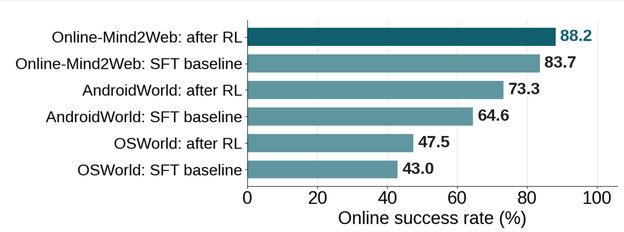


Table 1, redrawn: single reported values; "-" (unavailable) left blank; only UI-TARS-1.5 and -2 labelled.

Above UI-TARS-1.5 on every reported benchmark

SO WHAT → *Baseline step budgets and seeds are not stated in the text. UI-TARS-2 may use GUI or SDK actions.*

Same pipeline: RL vs SFT baseline, online (%)



Section 3.2: SFT baseline of the final iteration vs the final RL model.

SO WHAT → *RL ahead on all three; OSWorld and AndroidWorld were not the main training focus. PPO vs GRPO: the paper reports PPO consistently better in preliminary tests and shows training-reward curves only. Manual inspection found no substantial reward hacking, but the ORM has a relatively high false-positive rate.*

Headline Numbers

88.2%
Online-Mind2Web success
vs 83.7% SFT baseline

- 47.5% OSWorld
- 73.3% AndroidWorld
- 50.6% WAA (Windows)

The paper's own controls

CONTROL	UI-TARS-2	COMPARED WITH
BrowseComp-zh, SDK vs GUI	50.5 (SDK)	GUI-only 32.1
BrowseComp-en, SDK vs GUI	29.6 (SDK)	GUI-only 7.0
OSWorld, W4A8 quantized	44.4	47.5 full precision

Takeaway

One native model with a hybrid GUI + SDK action space reports 47.5 / 73.3 / 88.2 online on OSWorld / AndroidWorld / Online-Mind2Web, ahead of its own SFT baseline; seeds, matched budgets and cost are not reported.

59.8 mean score, 15 games | 45.3 Terminal Bench (SDK) | 68.7 SWE-Bench (SDK)

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Paper | Code